

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT
(25CS1302E)**

“CRM FOR SMALL BUSINESS”

A Project Report Submitted in the partial fulfillment of the requirements for the award of
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BACHELOR OF TECHNOLOGY

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Declaration

The Project Report entitled “CRM For Small Business” is a record of Bonafide work of **Chegireddy Thanvi – 2520030214, Garikapati Keerthi – 2520030416, P Sai Ashwika – 2520030224** submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering. The results embodied in this report have not been copied from any other departments/University/Institute

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CERTIFICATE

This is certified that the mini project-based report entitled “CRM For Small Business” is a Bonafide work of Chegireddy Thanvi – 2520030214, Garikapati Keerthi – 2520030416, P Sai Ashwika – 2520030224 in partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in Department of Computer Science Engineering, K L (Deemed to be University) during the academic year 2026-2027.

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Abstract

Small businesses need an efficient way to manage customer information and business activities. Manual records and spreadsheets can be difficult to maintain and may lead to duplication, missed follow-ups, and difficulty retrieving important information. The proposed CRM for Small Business provides a centralized platform for organizing customer-related information and sales activities.

The system is designed to store, search, update, and retrieve customer records in a structured manner. It includes Customer Management, Interaction Management, Lead Management, Sales Management, Follow-up Management, and a Dashboard. These modules provide an organized way to handle customer information and monitor business activities.

The project specifies HTML, CSS, and JavaScript for the frontend, Java / Node.js for the backend, MySQL for the database, Visual Studio Code as the development tool, and MySQL Workbench for database management. The overall goal is to reduce manual work, improve organization of business data, and support better business decisions.

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CHAPTER 1

1.1 Background of the Project

Small businesses frequently maintain customer details, communication records, leads, sales information, and follow-up activities using spreadsheets or manual records. As the amount of information grows, maintaining these records becomes difficult. Duplicate data, missed follow-ups, and slow retrieval of information can affect day-to-day business operations.

A Customer Relationship Management (CRM) system provides a centralized environment for managing customer-related activities. The project “CRM for Small Business” is intended to organize customer information and sales processes in one system. The supplied project material identifies customer records, interactions, leads, sales activities, follow-ups, and a dashboard as the major areas of the application.

1.2 Problem Statement

Customer information is often maintained using spreadsheets or manual records. Data may be duplicated, customer follow-ups can be missed, important information may be difficult to find quickly, and sales and customer activities may not be centrally managed. Manual work also consumes time and may cause errors.

The project therefore addresses the need for a simple and centralized CRM system that organizes customer information and sales processes.

1.3 Scope of the Project

The scope is a CRM system for small businesses focused on centralized customer and business-activity management. It covers customer details, interactions, leads, sales activities, follow-ups, and dashboard information.

The system is intended to provide a structured database-backed environment instead of disconnected manual records. Future extensions may include advanced reports, automated notifications, customer communication, and additional business insights.

1.4 Objectives of the Project

1. Develop an easy-to-use CRM system.
2. Reduce manual customer-data management.
3. Organize customer information efficiently.
4. Improve customer relationship management.
5. Manage sales processes effectively.
6. Support better business decisions.

7. Provide easy search, update, and retrieval of records.
8. Track interactions, leads, sales, and follow-ups.
9. Provide a dashboard for monitoring customer and sales activities.

1.5 Importance of the Application

The application provides a centralized way to manage information that may otherwise be distributed across spreadsheets and manual records. Centralized records can make information easier to retrieve and update. Tracking interactions and follow-ups provides a structured way to maintain customer relationships.

For small businesses, a simple CRM can reduce repetitive manual work and provide a clearer view of customer and sales activities.

1.6 Target Users / Audience

The primary target users are small businesses and people responsible for managing customer relationships and sales activities. These may include business owners, sales staff, customer-support staff, or other employees who maintain customer and business records.

CHAPTER 2

2.1 Hardware Requirements

A standard development computer is sufficient for building and testing the application. A modern processor, at least 8 GB RAM, adequate storage, and a stable internet connection are suitable for running the IDE, database tools, browser, and backend environment.

Users require a desktop or laptop capable of running a modern web browser. Requirements may increase if the system is later expanded to handle larger datasets or many concurrent users.

2.2 Software Requirements

The supplied project specifies:

- Frontend: HTML, CSS, JavaScript
- Backend: Java / Node.js
- Database: MySQL
- Development Tool: Visual Studio Code
- Database Management: MySQL Workbench

A modern web browser is required for accessing the frontend.

2.3 Development Tools and Frameworks

Visual Studio Code is used as the development environment and MySQL Workbench is used for database creation and management. HTML, CSS, and JavaScript form the frontend layer, while Java / Node.js is specified for backend development. The exact backend implementation should match the team's actual code.

CHAPTER 3 – TECHNOLOGY STACK

3.1 Frontend

HTML provides page structure, CSS provides layout and visual presentation, and JavaScript provides client-side interaction, validation, search behavior, and communication with backend services.

3.2 Backend

Java / Node.js is the specified backend technology. The backend receives requests, validates data, performs application logic, communicates with MySQL, and returns responses to the frontend.

3.3 Database – MySQL

MySQL provides structured storage for customer information and related business records. The database can store customer details, interactions, leads, sales activities, and follow-up information using related tables.

3.4 Development and Database Tools

Visual Studio Code supports application development. MySQL Workbench supports database design, SQL queries, table inspection, and database management.

CHAPTER 4 – SYSTEM ARCHITECTURE

4.1 High-Level Architecture

USER / BUSINESS STAFF Accesses the CRM application
FRONTEND HTML + CSS + JavaScript Forms • Search • Tables • Dashboard
BACKEND / APPLICATION LOGIC Java / Node.js Validation • Business Logic • CRUD
DATABASE MySQL Customers • Interactions • Leads • Sales • Follow-ups
RESULT Updated records, search results and dashboard information

Data flow: User → Frontend → Backend → MySQL Database → Backend → Frontend → User.

4.2 Description of Each Layer

Frontend Layer: provides forms, tables, navigation, search, and dashboard views.

Backend Layer: processes requests, validates input, executes application logic, and performs database operations.

Database Layer: stores persistent customer and business records.

Dashboard Layer: retrieves relevant data and presents a summarized view of customer and sales activity.

4.3 Data Flow

When a user performs an operation such as adding or searching for a customer, the frontend collects the input and sends it to the backend. The backend validates the request and communicates with MySQL. The database performs the required insert, update, search, or retrieval operation. The result is returned to the frontend for display.

CHAPTER 5 – DESIGN

5.1 Module Design

The CRM is organized into six modules identified in the project review:

1. Customer Management – add, update, search and manage customer details.
2. Interaction Management – track customer interactions and communication.
3. Lead Management – store and manage potential customers.
4. Sales Management – monitor sales activities and records.
5. Follow-up Management – schedule and track customer follow-ups.
6. Dashboard – display customer and sales activity information.

5.2 Database Design / ER Model

The following is a conceptual database model for documentation. Final table and column names should match the implemented database.

CUSTOMER	Customer_ID, Name, Contact, Email
INTERACTION	Interaction_ID, Customer_ID, Date, Type, Notes
LEAD	Lead_ID, Customer_ID, Status, Source
SALE	Sale_ID, Customer_ID, Amount, Date, Status
FOLLOW_UP	FollowUp_ID, Customer_ID, Date, Status, Notes
DASHBOARD	Aggregated customer and sales activity

Relationship concept: Customer is the central entity. Interactions, leads, sales, and follow-ups are associated with customer records using Customer_ID. Dashboard information is derived from stored customer and sales activities.

5.3 User Flow

A typical workflow begins at the CRM dashboard. The user can add or search for a customer, view customer information, record an interaction, manage a lead, record a sales activity, or schedule a follow-up. Updated information is stored in the database and can be retrieved later.

CHAPTER 6 – IMPLEMENTATION

6.1 Module-Wise Implementation

Customer Management provides operations for creating, viewing, updating, and searching customer records.

Interaction Management records communication or interactions associated with customers.

Lead Management stores potential customer information and tracks lead-related status.

Sales Management stores and monitors sales activities and related customer information.

Follow-up Management stores follow-up details and supports tracking of scheduled or completed follow-ups.

Dashboard combines relevant information into a summary view of customer and sales activity.

6.2 Frontend Logic

The frontend provides forms for entering customer and activity information and views for displaying stored data. JavaScript can validate input, handle user actions, perform searches, and communicate with backend APIs. The interface should provide clear navigation between customer, interaction, lead, sales, follow-up, and dashboard areas.

6.3 Backend Logic

The backend receives requests from the frontend and performs application operations such as creating customers, retrieving records, updating customer information, recording interactions, managing leads, recording sales, and maintaining follow-up information.

6.4 Database Operations

CRUD operations are central to the CRM. Create inserts new records, Read retrieves records, Update modifies existing information, and Delete—if implemented—removes records according to application rules. MySQL Workbench can be used to inspect tables and execute SQL during development.

CHAPTER 7 – FEATURES

7.1 Main Features

- Centralized customer information management.
- Add, update, search, and retrieve customer records.
- Customer interaction tracking.
- Lead management.
- Sales activity management.
- Follow-up scheduling and tracking.
- Dashboard for customer and sales activity.
- Structured MySQL database storage.
- Web-based interface using HTML, CSS, and JavaScript.

7.2 How Each Feature Works

Customer Management: user enters customer details and submits them; the backend validates and stores the record.

Interaction Management: user records communication associated with a customer.

Lead Management: potential customer information is recorded and maintained.

Sales Management: sales records are stored and associated with customer information.

Follow-up Management: follow-up information is recorded and tracked.

Dashboard: stored data is retrieved and summarized for quick monitoring.

CHAPTER 8 – TESTING

8.1 Testing Approach

Testing focuses on CRUD operations, validation, communication between frontend and backend, database storage/retrieval, and application performance, matching the testing focus identified in the supplied project material.

8.2 CRUD Testing

Create: verify that a valid record is inserted. Read: verify that saved records can be retrieved. Update: verify that modifications persist. Delete: if implemented, verify that deletion follows the application's rules.

8.3 Validation Testing

Validation should check required fields, acceptable formats, and invalid or incomplete submissions. Invalid data should not be inserted into the database, and the user should receive a clear response.

8.4 Sample Test Cases and Results

Test Case	Input / Action	Expected Result	Status
TC01	Add valid customer	Customer record is stored	To be filled
TC02	Search existing customer	Correct details displayed	To be filled
TC03	Update customer	Updated information is stored	To be filled
TC04	Invalid required field	Validation message displayed	To be filled
TC05	Add interaction	Interaction linked to customer	To be filled
TC06	Add lead	Lead information stored	To be filled
TC07	Record sale	Sales record stored	To be filled

TC08	Schedule follow-up	Follow-up stored/tracked	To be filled
TC09	Open dashboard	Customer/sales activity displayed	To be filled

Note: The supplied files describe the testing focus but do not provide executed test results, so the Status column is intentionally left for the team to complete.

CHAPTER 9 – DEPLOYMENT

9.1 Steps to Deploy

1. Install the required development/runtime environment.
2. Install and configure MySQL.
3. Create the CRM database and required tables.
4. Configure backend database connection details.
5. Start the backend application.
6. Start/serve the frontend.
7. Open the application in a modern browser.
8. Verify database connectivity and CRUD operations.

9.2 Environment Configuration

Backend configuration requires database connection details such as host, port, database name, username, and password. These values should be managed securely rather than hard-coded. The frontend should point to the correct backend/API address.

9.3 Hosting of Frontend and Backend

The supplied project material does not identify a final production hosting provider. A production deployment should consider secure hosting, database backups, availability, scalability, and expected user traffic.

CHAPTER 10 – CHALLENGES AND LIMITATIONS

10.1 Issues Faced During Development

Potential development challenges include organizing customer information into a suitable database structure, maintaining relationships between customers and activities, validating user input, and synchronizing frontend, backend, and database operations.

10.2 Solutions Applied

A modular design separates customer, interaction, lead, sales, follow-up, and dashboard responsibilities. MySQL provides structured storage, while CRUD operations provide a consistent data-management approach. Frontend and backend validation can reduce invalid data.

10.3 Current Limitations / Known Gaps

The supplied project files do not provide a complete list of confirmed bugs or implementation limitations. Therefore, this sample report does not claim specific bugs as existing facts. The review material identifies advanced reporting, automated follow-up notifications, customer communication, and more dashboard insights as future scope.

CHAPTER 11 – FUTURE ENHANCEMENTS

11.1 Planned Features

The project review identifies these future enhancements:

1. Advanced sales reports and analytics.
2. Automated notifications for follow-ups.
3. Customer communication features.
4. More dashboard and business insights.

11.2 Possible Integrations or Optimizations

Future versions can add richer reporting, automated reminders, communication tools, role-based access, and additional dashboard metrics. Database indexing and query optimization can be considered as the amount of business data grows.

CHAPTER 12 – CONCLUSION

12.1 Summary of the Project

CRM for Small Business is designed to provide a centralized platform for managing customer information and related business activities. It addresses problems associated with manual records and spreadsheets by organizing customer, interaction, lead, sales, and follow-up information in a database-backed application.

12.2 What Was Achieved

The project defines a CRM solution with six major modules: Customer Management, Interaction Management, Lead Management, Sales Management, Follow-up Management, and Dashboard. It also defines a technology stack consisting of HTML, CSS, JavaScript, Java / Node.js, and MySQL, supported by Visual Studio Code and MySQL Workbench.

12.3 Skills Learned During Development

The project provides exposure to requirement analysis, frontend development, backend logic, relational database design, CRUD operations, database management, system architecture, testing, documentation, and project presentation.

CHAPTER 13 – REFERENCES

1. Salesforce, “CRM for Small Business: Software & Tools.”
2. Oracle MySQL, “MySQL 8.0 Reference Manual.”
3. PROJECT REVIEW 1 TEMPLATE – CRM for Small Business, Team 7, Section S9.
4. PBL Project Poster Template – CRM for Small Business.
5. SAMPLE Project Documentation – supplied project-report reference.

CHAPTER 14 – APPENDICES

Appendix A – Suggested Screenshots

- Dashboard
- Customer Management
- Add/Update Customer
- Interaction Management
- Lead Management
- Sales Management
- Follow-up Management
- MySQL Workbench tables
- API/Postman testing (if applicable)

Appendix B – Sample SQL Structure

```
CREATE TABLE Customer (  
customer_id INT PRIMARY KEY AUTO_INCREMENT,  
name VARCHAR(100) NOT NULL,  
email VARCHAR(120),  
phone VARCHAR(20)  
);
```

```
CREATE TABLE Interaction (  
interaction_id INT PRIMARY KEY AUTO_INCREMENT,  
customer_id INT,  
interaction_date DATE,  
interaction_type VARCHAR(50),  
notes TEXT,  
FOREIGN KEY (customer_id) REFERENCES Customer(customer_id)  
);
```

Appendix C – User Guide

1. Open the CRM application.
2. Add customer details.

3. Search or update customer records.
4. Record customer interactions.
5. Add and manage leads.
6. Record sales activities.
7. Create and track follow-ups.
8. Use the dashboard to monitor customer and sales activity.

Appendix D – Team Contribution

Team Member	Role / Contribution
G Keerthi	Abstract, Introduction, Existing System part
P Sai Ashwika	Problem, Solution, Objectives
C Thanvi	Modules, Software Requirements, Conclusion